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Serial Port, Serial FIFO Board and MIDI of the G2 Amiga

Paula's serial port and a word-wide serial FIFO board, carried by an FTDI USB serial adapter, and MIDI through any class-compliant USB-MIDI interface, all bridged by the firmware on the G2's hub.

HIGHLIGHTS

- Paula's UART on an FTDI adapter: 0 lost at 115,200 baud.
- Serial FIFO board with g2serial.device: 0 lost to 500,000 baud.
- USB-MIDI on Paula's port: 0 lost at 100 % load.
- serialmon, midimon, lsz/lrz.

KEY PARAMETERS

PARAMETER	VALUE
Paula FIFOs	1,023 in / 255 out
FIFO board	to 500,000 baud
MIDI rate (NTSC)	31,126 baud
USB packets	64 bytes / ms

1 PATHS

PATH	AMIGA SIDE	USB SIDE	RATE
Paula's UART	serial.device (ROM) or g2serial.device on Paula	An FTDI serial adapter (FT232R and compatible) in serial mode	Paula's SERPER rate, or a fixed rate set by the console (`midi baud`)
Serial FIFO board	g2serial.device (Zorro II board, manufacturer 0x0A1C, product 1): words on the bus, no per-byte interrupt	The same FTDI adapter	Set by the driver; 0 lost to 500,000 baud
MIDI	Paula's serial port at the MIDI rate (NTSC SERPER 114: 31,126 baud)	A class-compliant USB-MIDI interface, cable 0	Event packets both ways
MPSSE	A byte stream through g2serial.device	FT2232H channel A in MPSSE mode (FTDI=MPSSE)	About 14,000 bytes/s each way

The bridge runs Paula's UART on the core's own clock: its transmitter is held while the bytes have nowhere to go and a byte reaches its receiver only when the receive buffer is empty, so no byte is lost to the clock gate. FIFOs: 1,023 bytes from Paula, 255 bytes to Paula; one 64-byte USB packet a millisecond each way.

2 MEASURED LOOPBACK (SERIALMON, 10 S AT 100 % LOAD)

DRIVER, PATH	BAUD	LOST	BYTES/S	ROUND TRIP MEAN
serial.device (ROM)	57,600	0	5,624	32.8 ms
serial.device (ROM)	115,200	0	10,453	28.8 ms
g2serial.device, Paula	115,200	0	11,138	29.5 ms
g2serial.device, FIFO board	115,200	0	11,476	10.4 ms
g2serial.device, FIFO board (FT232R, 5 s)	500,000	0	about 27,000	—

Errors, overruns and parity errors 0 in every run listed. Above 500,000 baud the FIFO path loses bytes (576,000: 14); the adapter's RS-232 transceiver is not measured. Round-trip figures read up to 19 % high (taken with a PAL line clock on an NTSC machine).

3 MIDI

ITEM	VALUE
Device	Any class-compliant USB-MIDI interface on a hub port, or an FTDI adapter at a fixed 31,250 bit/s
Transport	Both endpoints on the EHCI periodic schedule with interrupt on completion, served in the interrupt handler
Measured	midimon loopback through an M-Audio 0763:1050 DIN loop: three 10 s runs at 100 % load, 0 lost, 0 errors; round trip 300–560 ms (the device's buffering). FT232R at 31,126 bit/s, 100 %: 9,356 of 9,356, 41.9 ms mean
Console	`midi max` (4–64 bytes an OUT transfer), `midi reset`, `midi baud`, `midi trace on|off|save`

4 PROGRAMS

PROGRAM	INSTALL	FUNCTION
serialmon	C:	Loopback test, meters, modem lines, monitor, driver switching; ZMODEM, YMODEM, XMODEM through lsz/lrz; VT100/VT220 terminal
midimon	C:	MIDI monitor and loopback tester at 31,250 baud, message sender
g2serial.device	DEVS:	serial.device command set on the FIFO board or Paula
lsz, lrz	C:	lrzsz 0.12.20 senders and receivers (GPL-2.0), started by serialmon

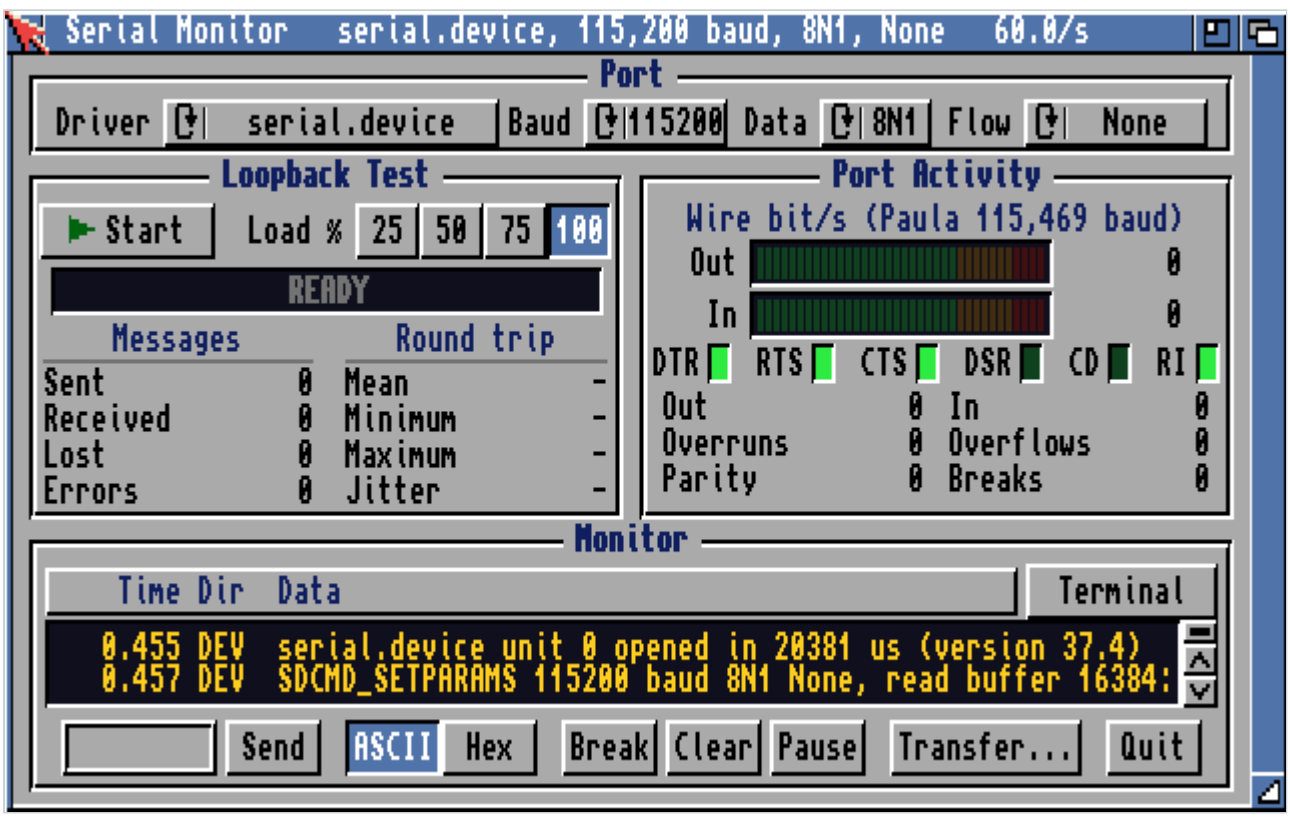


Figure 1. serialmon on a G2 revision C unit: NTSC 640 × 200 Workbench screen, captured with the firmware's snap.

5 LIMITS AND UNTESTED ITEMS

ITEM	STATE
Transfers to a far end over a cable	Untested: the unit has one serial adapter (Paula and the FIFO board share it)
USB MIDI on fabric 0x0AB5EF2D	Not repeated since the Paula receive holding register changed the receive path
Paula receive overruns at full MIDI load	11 in 91,007 bytes (an interrupt above level 5 suspected)

Measured on a G2 revision C unit with firmware release 1.0 and fabric 0x0AB5EF2D unless stated. Repository mikewolak/pano-g2-amiga, release 1.0.